

WASP-39b

R-1157

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-39_230531 · created 2026-07-31T01:56:22+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460095.7487 +/- 0.0046 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.183 +/- 0.016
Transit depth (Rp/Rs)^2	3.34 +/- 0.57 [%]
Orbital Inclination (inc)	87.81 +/- 0.85
Airmass coefficient 1 (a1)	1.396 +/- 0.021
Airmass coefficient 2 (a2)	-0.262 +/- 0.08
Scatter in the residuals of the lightcurve fit is	1.51 %
Best Comparison Star	#3 - [39, 430]
Optimal Method	PSF photometry
Transit Duration (day)	0.103 +/- 0.0043

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.183 ± 0.016 vs 0.1457 (deviation 2.23σ)
Duration fit vs published	0.103 d vs 0.12 d (deviation 3.78σ)
Mid-time O–C	-13.5 min
Depth SNR	10.9
Residual scatter	1.51 %

Light curve

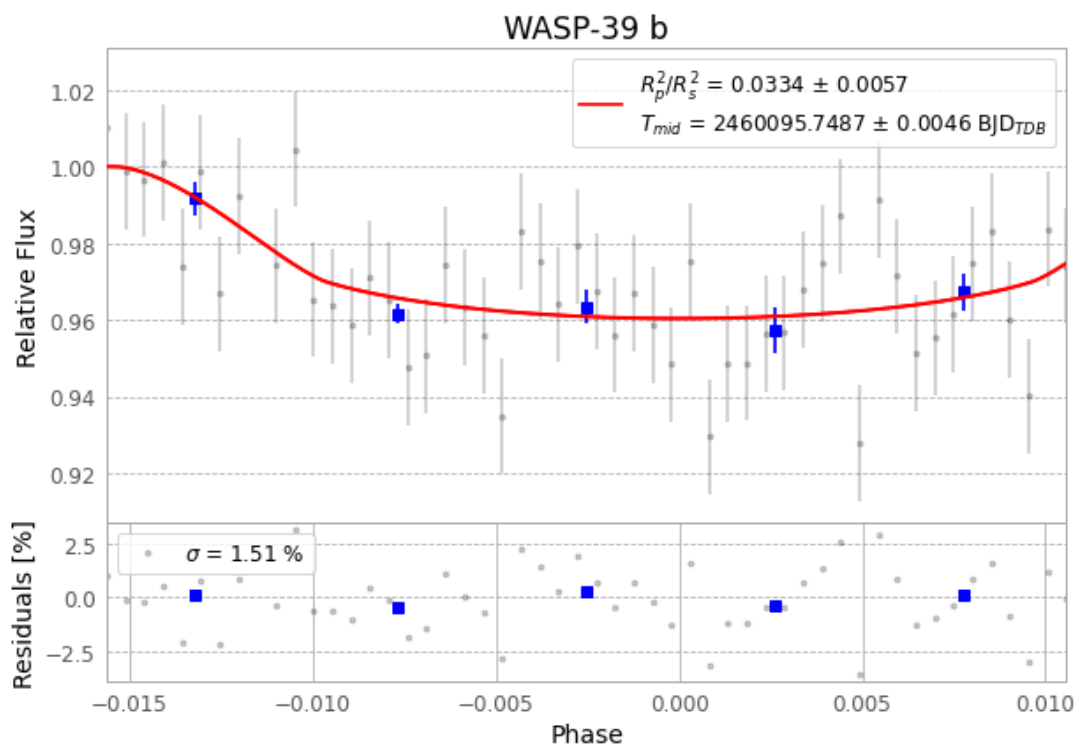


Figure 1 — FinalLightCurve_WASP-39 b_2023-05-30.png (SHA-256 825cb8e65442f35d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [127, 298], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d26ece1cb014ae3d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

52 FITS frames (34 MB), WASP-39230531041816.FITS → WASP-39230531065116.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-230531.log (39cdfa14f553...), FinalLightCurve_WASP-39 b_2023-05-30.png (825cb8e65442...), FinalLightCurve_WASP-39 b_2023-05-30.csv (23a1f71f816a...)

Compute resources

Wall time 170.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 01:56Z.